MarginOS Spec Amendment v16.1: answers to the developer's open questions

For: Design & Development Agent (handover) and Akthar Uddin (decisions)
Amends: MarginOS Product Spec v16 (`/workspace/spec.md`, now section 15 and the section 14 log)
Date: 29 October 2026
Status: Part B is ready to use. Part C needs Akthar's approval or action; until he decides, v16 stays as written.

*Answers to the five open questions the Design & Development Agent logged in docs/OPEN-QUESTIONS.md after building the v16 launch slice. Prepared by the Business & Product Agent on 29 October 2026. All web sources were checked on **29 October 2026**. This amendment does **not** change pricing, positioning or features. Where it clarifies sections 6, 12.0, 12.3, 12.7, 12.8, 12.10 or 5, the clarification applies. Anything that would change behaviour is listed under "Needs Akthar's approval or action" and stays as written in v16 until he decides.*

A. Summary

Q	Topic	Ready for the developer now	Needs Akthar's approval or action
Q1	CSV columns	Column lists, parsing rules and a column → MarginOS field mapping for every file (A1)	Ask friendly sellers for sample exports (K1). Optionally approve auto-detecting Etsy digital listings (K2)
Q2	Fee presets	Fee-type mapping clarifications and how the presets are applied (A2)	**Sign off the DRAFT fee preset table** (K3), including which Shopify plan the preset uses and how Amazon referral fees are compared
Q3	Tax rate sources	Official sources, how to store rates, and when to recheck them (A3)	None
Q4	Live Stripe Managed Payments	Developer checklist: keys, webhooks, test mode, then live (A4). It can start once Akthar's Stripe account exists	Akthar's own Stripe tasks: account, identity and business checks, bank, eligibility review, tax-inclusive prices, the approved fixed prices (K4)
Q5	Real email / Google sign-in	Security and verification rules that apply whatever tool is used (A5)	**Choose the sign-in approach and the email-sending service** (K5). Set up the Google Cloud sign-in project (K6)

B. Answers ready to give the developer

A1. Q1: CSV columns and mapping (build against these; confirm against real exports)

General parsing rules (all platforms)

- Find each column by its **header name**, not by position. Ignore case and surrounding spaces, and treat a UTF-8 byte-order mark as part of the file, not the header. Columns that aren't recognised are ignored. Platforms add columns without notice: for example, Shopify's own help page says "Lineitem SKU", but real exports say "Lineitem sku", and Shopify Payments added "Payout ID" in January 2026.
- If a **required** column is missing, the import fails and the message names the column. If an **optional** column is missing, the import continues and the dependent figure is labelled "estimated" or "not available" (section 6 rules).
- Amounts can include currency signs and thousands separators, and some EU files use a comma as the decimal mark. Etsy statements write empty amounts as "--". Strip the symbols and parse using the file's locale. "--" means empty, not zero.
- Dates come in several formats: text like "September 3, 2026" or "16 January, 2024", ISO dates with a time-zone offset, and Amazon's local formats. Parse all of them, and record the time zone if the file gives one.
- **Amazon's file starts with several description lines before the header row.** Find the header row by looking for its first columns ("date/time", "settlement id"), not by assuming it is row 1. Amazon.de, .fr, .it and .es reports can have **translated headers** (for example "Datum/Uhrzeit", "Abrechnungsnummer", "Bestellnummer", "Umsätze", "Verkaufsgebühren"). The Etsy Monthly Statement can also be translated (German "Datum, Art, Titel..."). Keep one header dictionary per language, and reject an unknown language with a clear message.

- Personal data columns are dropped at import, as section 6 requires. In the mapping tables below they are marked **strip**.

1. Shopify Orders export (Orders → Export; one row per line item, with order-level fields blank on later lines). Source: Shopify Help: Exporting orders. The same column set is used in every country; the currency is the store's base currency.

Column (as documented)	MarginOS field
Name (order number, e.g. #1001)	Order reference shown to the user
Id	Order ID (match key, 12.7)
Created at	Order date (period)
Paid at, Fulfilled at, Canceled at	Status dates. A cancelled order (Canceled at set) with Financial Status "voided" or "refunded" is excluded from revenue, and the refund is counted
Financial Status, Fulfillment Status	Order status
Currency	Order currency (converted at the ECB rate, section 6)
Subtotal, Shipping, Taxes, Total	Order totals. **Shipping = shipping charged**, not the seller's cost (section 6). Taxes = tax charged
Discount Code, Discount Amount	Order-level discount (split across lines by value, 12.0)
Lineitem quantity, Lineitem name, Lineitem price, Lineitem sku	Line units, product name, unit price, SKU (product match key)
Lineitem compare-at price	Not used
Lineitem discount	Line-level discount
Lineitem requires shipping	Physical (true) or digital (false) line (12.0)
Lineitem taxable, Tax 1–5 Name / Value	Tax detail per line, where present
Lineitem fulfillment status	Not used
Refunded Amount	Order-level refund (split by line value, 12.0; units estimated, 12.0)
Shipping Country, Shipping Province	Destination country and region (US state, CA province, AU state) for tax. **Billing Country** is the fallback
Payment Method, Payment ID, Payment References	Payment method, plus the link to the Shopify Payments export
Source, Tags	Source (web, pos, draft). Tags containing "test" are flagged, not dropped silently
Vendor, Risk Level, Payment terms, Next payment due at, Checkout fees, Outstanding Balance, Employee, Location, Device ID, Notes, Note Attributes, Accepts Marketing	Not used. **Notes / Note Attributes: strip**
Email, Phone, Billing / Shipping Name, Street, Address1–2, Company, City, Zip, Province Name, Phone	**Strip** (keep only the country and province code)

2. Shopify Payments transactions export (Finance → Payouts → View transactions → Export; amounts in the payout currency). Source: Shopify Help: Viewing and exporting payout details. Shopify documents the first 11 columns. The remaining ones are seen in real exports, so **confirm them against a real export**.

Column	MarginOS field
Transaction Date	Fee date
Type (charge, refund, dispute, adjustment, payout, reserve, ...)	Row kind. Only charge/refund/dispute rows are used for fees. **Skip payout and reserve rows**. Show adjustment rows in "Show the maths" without counting them as compared fees
Order	Join to the Orders export (by Name)
Amount, Fee, Net	Gross amount, **actual payment processing fee**, and net. Net = Amount − Fee
Currency	Payout currency
Presentment Amount, Presentment Currency *(confirm)*	What the buyer paid, in their currency

Column	MarginOS field
Card Brand, Card Source, Payout Status, Payout Date, Available On, Checkout, Payment Method Name, Payout ID *(confirm)*	Not used, except Payment Method Name (shown in "Show the maths")

- **Conversion fees (clarifies section 6):** the export has one "Fee" column. Shopify's international and currency-conversion charges are part of it and are **not given separately**. Unless a real export shows a separate column, record the whole Fee as **payment processing**, and say in "Show the maths" that it may include international and conversion charges. Don't invent a split.
- Rows can't be matched to a unique ID: use the full-row hash rule (12.7).

3. Etsy Orders CSV and Order Items CSV (Shop Manager → Settings → Options → Download Data → Orders → CSV type "Orders" or "Order Items", month and year). Source: Etsy Help: Download a spreadsheet of your sold transactions. Etsy doesn't publish a column list, so the lists below come from real files described by third parties. **Confirm against a real export.**

Orders CSV column	MarginOS field
Order ID	Order ID (join key)
Sale Date	Order date
Currency	Order currency
Order Value, Discount Amount, Shipping Discount, Shipping, Sales Tax, Order Total	Order totals. **Shipping = shipping charged**
Card Processing Fees	**Actual** payment processing fee (section 6)
Adjusted Order Total, Adjusted Card Processing Fees, Adjusted Net Order Amount	After refunds: refund = Order Total − Adjusted Order Total. Use the adjusted fee as the actual fee
Order Net	Check figure only
Status, Date Shipped, Payment Method, Payment Type, Order Type	Status and type (Order Type separates online from in-person sales)
Ship Country, Ship State	Destination country and region
Number of Items, Coupon Code, Coupon Details, InPerson Discount, InPerson Location, SKU	Not used, except SKU as a fallback
Buyer, Buyer User ID, Full Name, First Name, Last Name, Street 1–2, Ship City, Ship Zipcode	**Strip**

Order Items CSV column	MarginOS field
Order ID, Transaction ID	Match key (12.7)
Listing ID, SKU, Item Name, Variations	Product match: SKU first, then listing ID + variation, then title (12.7)
Quantity, Price, Item Total, Discount Amount, Shipping Discount	Line units, price and discount
Order Shipping, Order Sales Tax	Order-level figures repeated on each line. **Count them once per order**, then split by line value
VAT Paid by Buyer	Tax detail (UK/EU)
Currency, Sale Date, Date Paid, Date Shipped	Currency and dates
Listings Type	Possibly marks digital listings. **Not used automatically at launch** (see K2)
Order Type, Payment Type, InPerson fields, Coupon fields	Not used
Buyer, Ship Name, Ship Address1–2, Ship City, Ship Zipcode	**Strip** (keep Ship Country and Ship State)

4. Etsy Monthly Statement CSV (Shop Manager → Finances → Monthly statements → Download CSV; optional). Each activity is one row, in the **payment account currency**. The columns are **Date, Type, Title, Info, Currency, Amount, Fees & Taxes, Net, Tax Details**. Some 2026 files also have **Status** and **Availability Date** headings. **Confirm against a real export.**

Row (Type + Title)	MarginOS fee type
Sale "Payment for Order #..."	Check figure only (sales come from the Orders CSV)
Fee "Transaction fee: ..." and "Transaction fee: Shipping"	Referral / commission (actual)
Fee "Processing fee"	Payment processing (actual)
Fee "Listing fee", "Listing fee (auto-renew sold)", "Renew sold"	Listing (actual)
Fee "Regulatory operating fee"	**Referral / commission** (actual; see A2)
Fee "Credit for ..."	A negative of the matching fee type
Marketing "Etsy Ads", "Offsite Ads fee"	Ad spend (not compared)
Shipping "... postage label" / "shipping label"	Shipping cost for the order in Info (not compared)
Tax / VAT "VAT: seller fees", "Sales tax paid by buyer"	VAT on fees follows 12.8. Buyer tax is not seller revenue
Fee "Currency conversion"	Conversion fee (Pro cost line, section 6)
Refund "Refund to buyer for Order #..."	Check figure only (refunds come from the Orders CSV)
Deposit "... sent to your bank account"	Skip
Any other Type/Title	Counted as "other marketplace fees" (not compared) and listed in "Show the maths" for review

- The **Info** column links a row to its order ("Order #...") or listing ("Listing #..."). Monthly fees with no order (Etsy Ads, VAT on fees, subscriptions) are store-level costs for the month.

5. Amazon Seller Central: Payments → Reports Repository → Transaction (date range) report, "Custom Unified Transaction" CSV. Amazon's help text says "columns in the report might vary by country". Amazon's help pages need a sign-in, so these column names come from Amazon's help text quoted by a third party and from sellers' real files. **Confirm against a real export from each marketplace.**

Column (US file; UK and EU variants noted)	MarginOS field
date/time	Posting date (period, section 6)
settlement id	Not used (it is part of the row hash)
type (Order, Refund, Service Fee, Adjustment, Transfer, FBA Inventory Fee, Shipping Services, ...)	Row kind. **Transfer rows are skipped**. Unknown types are counted as "other marketplace fees" and flagged
order id	Order ID
sku, description, quantity	SKU, name, units
marketplace (e.g. amazon.com, amazon.co.uk, amazon.de)	**Region of the order** for presets and tax
account type, tax collection model	Tax collection model: marketplace-collected tax (12.8)
fulfillment / **fulfilment** (UK spelling)	FBA ("Amazon") or seller-fulfilled ("Seller")
order city, order postal	**Strip**
order state	Region (US state, CA province, AU state)
product sales, product sales tax	Product revenue and the tax on it
shipping credits / **postage credits** (UK), shipping credits tax	Shipping charged and its tax
gift wrap credits, giftwrap credits tax	Revenue (gift wrap) and its tax
promotional rebates, promotional rebates tax	Discount
marketplace withheld tax	Marketplace-collected tax (not the seller's liability, 12.8)
Regulatory Fee, Tax On Regulatory Fee	**Referral / commission** (A2). Tax on fees follows 12.8
selling fees	**Referral / commission (actual)**. This includes any closing fee
fba fees	Fulfilment (counted, not compared)

Column (US file; UK and EU variants noted)	MarginOS field
other transaction fees	Other marketplace fees (counted, not compared)
other	Amounts on Service Fee rows: subscription → overhead; "Cost of Advertising" → ad spend; FBA storage → fulfilment. Anything else → other marketplace fees
total	Check figure: the sum of the money columns must equal total on each row
Transaction Status, Transaction Release Date (added in some accounts in 2026)	Deferred rows: count them on their **posting date** like any other row. Never count a row twice when its status changes; the full-row hash plus status handles this

- Amazon's report has **no separate payment processing fee**: payment processing is part of "selling fees". So Amazon only has one compared fee type (referral / commission).

Getting real samples (recommended). Before writing the final parsers, the developer should have **at least one real export of every file above**, ideally one each from UK, US and an EU marketplace, and the Amazon report in German or French as well. Friendly-seller testing is already planned, so ask those sellers (see K1). To protect buyers' privacy, store only the **header row plus a few rows with buyer columns blanked** as test fixtures. Until real samples are checked, every column marked "confirm" above stays a working assumption.

A2. Q2: How the fee presets are applied (clarifies 12.3)

- **Fee-type mapping additions (clarifications, no behaviour change):**
  - Etsy "Regulatory operating fee" and Amazon "Regulatory Fee" are platform charges on each sale, so they map to **Referral / commission** and are included in that preset where the seller's country charges them.
  - A new row in the 12.3 mapping table: **Other marketplace fees** (Amazon "other transaction fees", unknown statement fees). They are **counted in fees, not compared**.
  - Etsy currency conversion (2.5%) is a **conversion fee** (Pro cost line), not a compared fee.
  - Shopify plan fees, Amazon selling-plan fees and Etsy Plus are **subscriptions**. They are overheads only if the user enters them (12.3, as now).
- **Region of a sale:**
  - Shopify: the store's country (from the user's settings, defaulting to the home currency's region).
  - Etsy: the country of the seller's payment account (Etsy charges processing fees by bank location).
  - Amazon: the "marketplace" column.
  - For **EU**: Etsy's processing fee is the same in all euro countries. The regulatory operating fee applies only in France, Italy and Spain (and Hungary per one Etsy page). Shopify's rates can differ by EU country. Store EU presets **per country** where they differ, with the EU row as the default.
- **Fixed fees published in another currency** (Etsy's $0.20 listing fee is published in USD) are converted at the ECB rate on the order date (section 6), and the conversion is shown in "Show the maths".
- **Listing fee preset:** $0.20 per unit sold (Etsy charges $0.20 per extra unit sold, and the original listing fee is already paid for the first). The actual listing fee in the Monthly Statement also includes fees for listings that didn't sell. That is a real cost and stays in the comparison.
- **Payment processing preset = the standard domestic online card rate.** Actual fees include higher rates for international, Amex and premium cards. That is a real cost to the seller, so a fee drag it causes is valid. "Show the maths" should say "your actual rate includes international and premium cards".
- **Amazon referral preset at launch:** 15% (Amazon's "Everything else" rate in every region, with that region's minimum per item), unless Akthar decides otherwise (K3b).
- The table is filled in from the DRAFT below. It goes live **only after Akthar's sign-off** (K3), and is then rechecked every 6 months (12.3).

DRAFT fee preset table: needs Akthar sign-off. All rates are the platforms' standard published rates, checked 29 October 2026, excluding VAT unless noted.

Shopify Payments, online standard card rate (payment processing), by plan. Plan prices are shown for information only (subscriptions aren't compared).

Region	Basic	Grow	Advanced	Plus	Amex / international (Basic)	Third-party gateway fee (Basic / Grow / Advanced / Plus)	Source
UK (GBP)	2% + 25p	1.7% + 25p	1.5% + 25p	1.3% + 25p	3.1% + 25p	2% / 1% / 0.6% / 0.2%	shopify.com/uk/pricing
EU (EUR, Ireland checked; excl. VAT)	2% + €0.25	1.7% + €0.25	1.5% + €0.25	1.4% + €0.25	3.1% + €0.25	2% / 1% / 0.6% / 0.2%	shopify.com/ie/pricing. **Other EU countries: confirm on each country's Shopify pricing page**
US (USD)	2.9% + 30¢	2.7% + 30¢	2.5% + 30¢	2.25% + 30¢	premium 3.5% + 30¢; international +1%	2% / 1% / 0.6% / 0.2%	shopify.com/pricing
Canada (CAD)	2.8% + 30¢	2.6% + 30¢	2.4% + 30¢	2.15% + 30¢	3.5% + 30¢	2% / 1% / 0.6% / 0.2%	shopify.com/ca/pricing
Australia (AUD)	1.7% + 30¢	1.55% + 30¢	1.35% + 30¢	1.15% + 30¢	Amex 2.9% + 30¢; international 3.5% + 30¢	2% / 1% / 0.6% / 0.2%	shopify.com/au/pricing

The third-party gateway fee applies when the seller doesn't use Shopify Payments. It is billed on the Shopify invoice and doesn't appear in any export. It is shown for reference only and isn't a preset at launch.

Etsy (the same for every shop unless noted). Sources: Fees and taxes for selling on Etsy, Payment processing fees, Regulatory operating fee, Fees & Payments Policy.

Region	Referral / commission: transaction fee	Referral / commission: regulatory operating fee	Payment processing	Listing
UK	6.5% of item + shipping + gift wrap	0.48%	4% + £0.20	$0.20 per unit sold (converted)
EU (euro countries)	6.5%	France 1.14%, Italy 0.80%, Spain 0.88%, Hungary 1.97% (listed on one Etsy page; confirm), others 0%	4% + €0.30 (non-euro EU countries such as Denmark and Sweden: 4% + a fixed fee in local currency; confirm)	$0.20 per unit sold
US	6.5%	0%	3% + $0.25	$0.20 per unit sold
Canada	6.5%	0.50%	3% + CA$0.25 for domestic or US orders; 4% + CA$0.25 for other international orders	$0.20 per unit sold
Australia	6.5%	0%	3% + A$0.25 domestic; 4% + A$0.25 international	$0.20 per unit sold

Not compared: Offsite Ads (15%, or 12% for shops with $10,000+ in sales over 12 months, capped at $100 per order) is **ad spend**. Currency conversion (2.5%) is a **conversion fee**. VAT on seller fees follows 12.8.

Amazon referral fees (referral / commission), main categories. FBA fees are **counted, not compared**. Selling-plan fees are subscriptions. Sources: US, UK, EU: Amazon.de (the other EU stores use the same European schedule for these categories; confirm each), Canada, Australia.

Category	US	UK	EU (Amazon.de)	Canada	Australia
Everything else (**proposed default preset**)	15%	15%	15%	15%	15%
Minimum per item	$0.30	£0.25	€0.30	CA$0.40	none
Clothing & accessories	5% ≤$15; 10% $15–20; 17% >$20	5% ≤£15; 10% £15–20; 15% >£20 (FBA/Prime >£40: 15% up to £40, 7% above)	5% ≤€15; 10% €15–20; 15% >€20 (Prime >€45: 15% up to €45, 7% above)	10% <CA$20; 17% ≥CA$20	10% ≤A$20; 13% >A$20
Home / kitchen	Home & Kitchen 15%	Home Products 8% ≤£20, 15% above; Kitchen 15%	Household 8% ≤€20, 15% above; Kitchen 15%	Home & Kitchen 15%	Home & Kitchen 13%
Beauty, health & personal care	8% ≤$10; 15% above	8% ≤£10; 15% above	8% ≤€10; 15% above	8% ≤CA$15; 15% above	8% ≤A$20; 13% above

Category	US	UK	EU (Amazon.de)	Canada	Australia
Jewellery	20% up to $250; 5% above	20% up to £225; 5% above	20% up to €250; 5% above	20% up to CA$325; 5% above	15% up to A$200; 5% above
Toys & games	15%	15%	15%	15%	12%
Consumer electronics	8%	7%	7%	8%	8%
Books & media	15% + $1.80 closing fee	15% (closing fee applies; confirm amount)	15%	15%	15% + A$1.00 closing fee
Handmade	not checked	12%	12%	not checked	not checked

The UK and EU clothing rates include Amazon's reductions effective 5 January 2026.

A3. Q3: Official tax rate sources and update cadence (clarifies 12.8)

Region	Official source	What to take from it
UK VAT	GOV.UK: VAT rates and HMRC: rates of VAT on different goods and services	20% standard, 5% reduced, 0% zero (standard rate unchanged since 4 January 2011)
EU VAT	European Commission: Taxes in Europe Database (TEDB) via the EC VAT rates page. Member states supply TEDB's data. Where TEDB and a national tax authority disagree, use the **national tax authority** and note it	Standard and reduced rates for each of the 27 countries. Several standard rates changed in 2024–2025, so don't copy an old list
Canada GST/HST	CRA: GST/HST calculator (and rates)	GST 5% (AB, BC, MB, QC, SK, NT, NU, YT); HST 13% ON, 14% NS (from 1 April 2025; 15% before), 15% NB, NL, PE. CRA's table also shows PST for information
Canada PST / QST / RST	The provincial governments' own tax pages (BC Ministry of Finance, Saskatchewan Finance, Manitoba Finance, Revenu Québec). The developer records each page's link when filling the table	BC 7%, MB 7%, SK 6%, QC 9.975% (as shown alongside CRA's GST/HST table on 29 October 2026; confirm on each provincial page)
Australia GST	ATO: GST and How GST works	10% on most goods and services (some GST-free)
US sales tax	**No built-in table** (12.8, unchanged): US tax comes only from the export. If a state's rate is ever needed for wording or support, the official sources are each **state's Department of Revenue** website and, for the 24 member states, the Streamlined Sales Tax rate and boundary files	Not used in calculations

- **Store rates with dates.** Each row has country (and province), rate type, rate, **valid from**, **valid to**, source link and the date checked. An order uses the rate **in force on its order date** (for example, a Nova Scotia order from March 2025 uses 15%, not 14%).
- **Rate per product:** at launch, use the **standard rate** unless the export gives the tax charged. Reduced and zero rates (for example UK children's clothes) are applied only from the export's tax amount, never guessed. "Show the maths" says which applied.
- **When to recheck:** at least every 6 months (12.8), plus after each **UK Budget** (usually autumn), on **1 January and 1 July** for the EU (when most rate changes take effect), and after **Canadian provincial budgets** (February–April). Australia: the 6-monthly check is enough. Record each check, even when nothing changed.
- These tables are **estimates, not tax advice** (12.8). The 12.8 instruction to recheck marketplace-collection guidance at build still applies.

A4. Q4: Live Stripe Managed Payments: the developer's checklist

The order matters. The developer can only start once Akthar has created the Stripe account and invited them (K4, steps 1–2). Sources: Managed Payments, Set up, Eligibility.

1. **Access:** use your own Stripe user, invited by Akthar with the Developer role. Never use a shared login. Use **sandbox / test-mode keys** first. Store keys only in the hosting provider's secret store, never in code or the repo. Use a **restricted key** with only the permissions the adapter needs (Checkout Sessions, Prices/Products read, Charges/Refunds, Disputes read, Events read).

2. **API version:** pin **2025-03-31.basil or later** (required by Stripe Managed Payments).
3. **Products and prices in test mode:** one product, "MarginOS Pro", with tax code **txcd_10103001** (Software as a service, business use; eligible for Managed Payments). It has two one-time prices, £59 and £39, each with **currency_options** USD 79 / 52, EUR 69 / 47, CAD 109 / 73, AUD 109 / 74. **tax_behavior = inclusive** must be set on every price and currency option. Keep the price IDs in configuration, not code.
4. **Checkout Session:** mode=payment, **managed_payments[enabled]=true**, client_reference_id = MarginOS account ID (also in metadata), expiry 30 minutes, success and cancel URLs. Create the session only after the allowed-country check (12.5).
5. **Webhook endpoint:** subscribe to exactly the events in 12.10, verify every signature with the endpoint's signing secret, process each event ID once, and read the latest Charge or Dispute from Stripe before deciding (12.10). Test locally with the Stripe CLI (`stripe listen --forward-to ...`) and Stripe's test cards (4242 4242 4242 4242).
6. **Run the full 12.10 test-mode list.** It must pass before going live. Also check in test mode that Adaptive Pricing shows for a currency without a fixed price, and that the receipt and statement descriptor are as expected.
7. **Ask Stripe** whether a Managed Payments account can block buyer countries on Stripe's side (12.5). Record the answer in the Open questions log.
8. **Going live (after Akthar confirms K4 is complete and Stripe has approved Managed Payments):**
    - Create the live product and prices exactly as in test, and check them against the approved price table (12.5).
    - Create a **separate live webhook endpoint** with its own secret, and switch to live keys.
    - Do **one real purchase and refund**. Akthar uses his own card; about £2.15 in fees is not returned (12.5).
    - Check that the licence record, tax country, early-bird counter and refund handling all work.
    - Switch on alerts for failed webhooks and the daily reconciliation (12.10).

A5. Q5: Sign-in rules that apply whatever tool is chosen (clarifies 5 and 10.2 criterion 1)

- **Email and password:**
    - Hash passwords with a modern slow hash: Argon2id, or bcrypt where Argon2id isn't available.
    - Passwords must be at least 8 characters, with **no other complexity rules**. Reject passwords found in known breaches (NIST SP 800-63B approach).
    - Rate-limit sign-in and reset attempts.
- **Email verification:**
    - The verification link is single-use and expires after 24 hours. A new link can be requested, rate-limited.
    - **Uploads and checkout stay locked until the email is verified** (12.4, 12.5, unchanged).
- **Google sign-in:**
    - Use Google's OpenID Connect sign-in and identify the user by Google's **"sub"** claim, never by email (Google OIDC reference).
    - Treat the email as verified **only when the ID token's "email_verified" is true**. Otherwise, send MarginOS's own verification email.
    - Validate the ID token's signature, issuer, audience and expiry.
- **Linking a Google sign-in to an existing email account (prevents account takeover):**
    - Link automatically only when Google says the email is verified.
    - If the existing email/password account was **never verified**, linking **removes its password**. The user can set a new one through a reset email. This stops someone from creating an unverified account with another person's email first.
- **Password reset:** the link is single-use and expires after 1 hour. A successful reset signs out all other sessions.
- **Sessions:** use secure, httpOnly, SameSite cookies. Sign the user out after inactivity in line with the hosting setup, and let users sign out of all devices.
- **Logs:** never log passwords, tokens or verification links. Account emails are account data (MarginOS is controller, 12.6), and each tool or service that stores them goes on the **subprocessor list** with a DPA (12.6).
- **Location:** the user database, sign-in service and email-sending service all store data in the **UK or EU** (12.6).
- **Tool-neutral design:** keep sign-in behind a small internal interface (sign up, sign in, verify, reset, link Google), like the payment layer, so the chosen tool can be replaced.

C. Needs Akthar's approval or action

K1. Ask friendly sellers for sample exports (Q1): action

To finish the parsers, the developer needs real files. Proposed request (a **draft for Akthar to send himself**; nothing has been sent):

> Hi [name], thanks again for helping test MarginOS. Could you send me one recent month of these exports from your shop? **Shopify:** Orders → Export (CSV), and Finance → Payouts → View transactions → Export. **Etsy:** Shop Manager → Settings → Options → Download Data → "Orders" and "Order Items" CSVs, plus that month's Finances → Monthly statement CSV. **Amazon:** Payments → Reports Repository → Transaction → custom date range CSV. Please blank out buyer names, emails and addresses first, or just send the first 20 rows. We only need the column layout and a few example rows. They will only be used to test the importer and will be deleted after testing. Thank you!

Aim for UK, US and at least one EU seller (an Amazon.de or Amazon.fr report would be ideal).

K2. Optional: auto-detect Etsy digital listings (Q1): approval

The Etsy Order Items CSV appears to have a **"Listings Type"** column. If a real export shows it marks digital listings, MarginOS could pre-mark those listings as digital, while the user can still change them. This would be a small change to how 12.0 works. **Default (no approval): keep v16.** Etsy orders are physical unless the user marks the listing or shop as digital.

K3. Sign off the DRAFT fee preset table (Q2): approval

a. **Approve the rates in the DRAFT table** (A2) as the launch presets, with the source links and the 29 October 2026 check date.

b. **Shopify plan:** the preset needs one plan per region. **Recommendation: use the Basic plan rate.** It is the most common plan for small sellers, and it never makes a Grow or Advanced seller's fees look worse than they are. A plan selector in settings would be a new feature that needs approval. Not recommended for launch.

c. **Amazon referral comparison:** Amazon's rate depends on the product category (5–20%), and the Transaction report doesn't include the category. The options: - (i) **Recommended: compare against 15%** (Amazon's default rate) and add a "Show the maths" note: "Amazon's referral rate depends on category (5–20%). If your category's rate is above 15%, part of this fee drag is your category's standard rate." - (ii) Let the user pick their main Amazon category (a new setting, so a feature change). - (iii) Don't compare Amazon referral fees at all. This would leave Amazon with no compared fee type, so the Fees part would never be used for Amazon-only stores. Not recommended.

d. **Other EU countries' Shopify rates** are to be confirmed by the developer from each country's Shopify page before the table goes live.

K4. Akthar's own Stripe Managed Payments tasks (Q4): action

Only the account owner can do these. Sources: Eligibility, Set up.

1. **Create the Stripe account** at dashboard.stripe.com for the MarginOS business (sole trader or limited company), with the business's legal name, UK address, the website URL and a description of the product ("software as a service for online sellers"). Switch on **two-step sign-in**.
2. **Invite the developer** with the Developer role. Don't share passwords or secret keys by message.
3. **Identity and business checks:**
    - Personal ID and home address.
    - For a limited company: the Companies House number, the directors, and anyone owning 25% or more.
    - The website must show the product, prices, terms, privacy policy, refund policy and a support contact before Stripe's review.
4. **Bank account:** a UK **GBP** account in the business's (or sole trader's) name, for payouts (12.5: GBP payouts only).

5. **Activate Managed Payments** in the Dashboard and **pass Stripe's eligibility review.** Stripe checks business type, location (UK is supported), that the product is digital and fully automated, and the dispute history. **Launch waits for Stripe's approval.**
6. **Tax settings:** set "Include tax in prices" so prices are **tax-inclusive** by default (12.5). The developer still sets it on each price.
7. **Prices:** confirm that the live prices match the **approved** table exactly: £59 / $79 / €69 / CA$109 / A$109, and early-bird £39 / $52 / €47 / CA$73 / A$74. **Recheck the exchange rates before launch.** If any price is more than 5% off its GBP equivalent, Business & Product flags it to Akthar (12.5; no change without his approval).
8. **Dashboard settings:**
    - Switch on **Adaptive Pricing**.
    - Set the **support email**, which must be answered within 48 hours.
    - Set the **statement descriptor** so statements read LINK.COM* MARGINOS (or the closest Stripe allows; tell Business & Product if it differs, because the checkout, receipt and FAQ wording depends on it).
9. **Before going live** (unchanged from v16): legal review of the terms and refund wording (12.5) and the accountant's check of MarginOS's VAT position (12.9).

K5. Choose the sign-in approach and email-sending service (Q5): approval

Any choice must meet: email verification, the Google "email_verified" rule, UK/EU data storage, a DPA, and listing on the subprocessor list (12.6). **Nothing has been chosen.** These are options with published prices (checked 29 October 2026; USD unless noted).

Option	What it is	Data location	Cost at MarginOS's likely scale	Fit
A. Open-source sign-in library inside the app (for example Better Auth or Auth.js; pick whichever suits the developer's framework)	Sign-in code runs in MarginOS's own app and database	Wherever MarginOS is hosted (UK/EU)	**£0 for the software.** Only the email-sending cost (below)	**Recommended.** No new account-data vendor, full control of the email_verified rule, nothing per user. Slightly more developer work than a hosted service
B. Supabase Auth (hosted)	Hosted sign-in and database	Region chosen per project, including **London (eu-west-2)** and Frankfurt	Free plan: 50,000 monthly active users, but free projects **pause after 1 week of inactivity** (not suitable for live use). **Pro from $25/month**, including 100,000 monthly active users	Good fallback if the developer prefers a hosted service. Needs a custom email sender for production emails
C. Auth0 (Okta)	Hosted sign-in	**UK or EU** tenant region available	**Free up to 25,000 monthly active users.** Paid (Essentials, consumer apps) from $35/month for 500 users, $70 for 1,000, $350 for 5,000	Works, but becomes expensive if paid features are needed
D. Clerk	Hosted sign-in	**US only** (EU/UK data transferred under the Data Privacy Framework)	Free up to 50,000 retained users; Pro $25/month	**Not compatible** with "store data in the UK or EU" (12.6) without a spec change. Not recommended
E. Keycloak (self-hosted)	Separate open-source sign-in server	Your own hosting	Software free; needs its own server and upkeep	Too much upkeep for a one-person launch. Not recommended

Email-sending service (needed for verification, password reset and the Free-plan reminder emails in 12.4 and 12.6):

Option	Data location	Cost
Amazon SES, London region	UK (eu-west-2)	$0.10 per 1,000 emails. New accounts may be placed on an "Essentials" plan at $0.16 per 1,000. A few pence a month at launch volumes
Brevo (French company)	EU (France and Germany)	Free plan: 300 emails per day. Paid plans if more are needed
Resend	Sends from Ireland, but **account data and logs are stored in the US**	Free: 3,000 per month (100 per day). Not recommended for the UK/EU rule

Recommendation for Akthar to approve:

- **Option A** (an open-source library in the app, £0) with **Amazon SES London** for email, both added to the subprocessor list.
- Fallback: **Option B** (Supabase Pro, London region, $25/month) with the same email sender.

Either choice needs Akthar's approval before the developer starts. Paid plans need his approval of the spend.

K6. Set up Google sign-in (Q5): action

Google sign-in needs a Google Cloud project owned by MarginOS, not by the developer's personal account:

- Create the project.
- Set up the OAuth consent screen (app name, support email, logo, homepage, privacy policy and terms links, and the authorised domain).
- Request only the basic scopes **openid, email, profile**. Google treats these as non-sensitive, so no security assessment is needed, but **brand verification can take a few days**.
- Add the developer as an editor.
- Create the web client ID for test and for live.

D. Sources (all checked 29 October 2026)

- Shopify: Exporting orders; Payout details and transaction export; pricing pages for the US, UK, Ireland/EU, Canada and Australia.
- Etsy: Download sold transactions CSV; Fees and taxes; Payment processing fees; Regulatory operating fee. Column lists were cross-checked against third-party descriptions of real files (Coupler.io, PeakApps Studio, Briefmatic). **These need confirming against a real export.**
- Amazon: pricing pages for the US, UK, DE, CA and AU. Transaction report columns come from Amazon's Date Range Reports help text (as quoted by LianLian), a seller's real header row posted on the SellersAskSellers forum, and Amazon Seller Forums posts on the 2026 report changes. Amazon's own help pages need a Seller Central sign-in. **These need confirming against a real export.**
- Tax: GOV.UK VAT rates; EC VAT rates and TEDB; CRA GST/HST rates; ATO How GST works (ato.gov.au blocked automated fetching; the 10% rate was confirmed from ATO search results); Streamlined Sales Tax.
- Stripe: Managed Payments; Set up; Eligibility.
- Sign-in and email: Supabase pricing and regions; Auth0 pricing and tenant regions; Clerk pricing and GDPR notice; Amazon SES pricing; Brevo plans and data location; Resend regions; Google OpenID Connect reference; Better Auth docs.

E. Open questions log entries (spec section 14)

ID	Date raised	Section	Question	Raised by	Answer	Answered by	Amendment
Q1	29 Oct 2026	6, 12.0, 12.7	Which exact columns do the Shopify (Orders, Payments transactions), Etsy (Orders, Order Items, Monthly Statement) and Amazon (Custom Unified Transaction) exports have, and how do they map to MarginOS fields?	Developer	Column lists, parsing rules and mappings are in v16.1 A1. Items marked "confirm" must be checked against real exports. Akthar to ask friendly sellers for sample files (K1). Optional Etsy digital auto-detect needs his approval (K2); default unchanged	Business & Product Agent	v16.1
Q2	29 Oct 2026	12.3	Can Business & Product supply and sign off the fee preset table (platform × region × compared fee type)?	Developer	DRAFT table with sources and check dates, plus mapping clarifications, in v16.1 A2. **Not live until Akthar signs off (K3)**, including the Shopify preset plan (Basic recommended) and the Amazon referral comparison (15% recommended)	Business & Product Agent (draft); Akthar (sign-off pending)	v16.1

ID	Date raised	Section	Question	Raised by	Answer	Answered by	Amendment
Q3	29 Oct 2026	12.8	Which official sources should the tax rate table use, and how often is it updated?	Developer	Sources, date-effective storage and recheck timing are in v16.1 A3. US unchanged: export only	Business & Product Agent	v16.1
Q4	29 Oct 2026	12.5, 12.10	What is needed to set up live Stripe Managed Payments?	Developer	Developer checklist in v16.1 A4. Akthar's account, verification, bank, eligibility and Dashboard tasks in K4 (action pending)	Business & Product Agent	v16.1
Q5	29 Oct 2026	5, 10.2, 12.6	Which real email/password and Google sign-in setup should be used?	Developer	Tool-neutral security rules in v16.1 A5. Options and costs in K5 (open-source library + Amazon SES London recommended; Supabase Pro fallback). **Choice and any spend need Akthar's approval**. Google Cloud setup is Akthar's action (K6)	Business & Product Agent (recommendation); Akthar (decision pending)	v16.1